Lecture notes

Unmatched discussion

Introduction to Data Mining

- **Goal**: Find useful patterns in data.
- The specific task depends on what is to be learned from the dataset.

Classification

- **Definition**: Learn from examples with known class labels to predict the label of a new example.
- **Example**: Classifying incoming emails using a collection of emails previously labeled as *spam* or *not spam*.

Sources: paragraph 1, paragraph 2

Clustering

- **Definition**: Grouping examples according to similarity when examples do not come with target class labels; groups are inferred from data rather than supplied as correct answers.
- **Example**: Grouping customers sharing similar purchasing behavior.
- **Key Qualifications and Limitations**:
    - Clustering outcomes depend on data representation and how similarity is measured.
    - A grouping can be mathematically convenient without being practically useful for the target application.

Sources: paragraph 3, paragraph 4, paragraph 5